

Cyber Security - Class 03 (Lab Class 2)

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Date: September 14, 2024

Topics Covered

1. MITRE ATT&CK Framework
 2. Cyber Kill Chain
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Key Notes

1. Useful Websites

- [Exposed.lol](#): Check if your email is available on the international black market.
- [HavelBeenPwned](#): Verify if your data has been compromised.

2. Enhanced Incident Detection with Threat Intelligence

- **MITRE ATT&CK Framework:**
 - A structured step-by-step procedure to track and respond to cyber threats.
 - **Key Components:**
 1. **Adversary:** Tracks how attackers gain access, move within networks, and steal data to develop defensive strategies.
 2. **Threat Models:** Includes gaining access and exploiting vulnerabilities.
 3. **Techniques:** Over 300 techniques for identifying threats.
 4. **Mitigating Tactics:** 18 techniques to decide defensive actions.
 - **Designed from an Attacker's Perspective.**

- **MITRE ATT&CK Matrices:**

1. Enterprise ATT&CK
2. Pre-ATT&CK
3. Mobile ATT&CK

- **Useful Tools:**

- [MITRE ATT&CK Navigator](#): Scanning IPs to find vulnerabilities.
- [MITRE ATT&CK Website](#): Explore matrices and ICS (Industrial Control Systems).

3. Command-Line Tools for Cybersecurity

- **Root Access:** `sudo -i`
- **Nmap:**
 - `nmap`: Find IP details.
 - `nmap --help`: Learn about attributes.
 - `nmap -sC -sV -Pn -A <target-ip>`: Scan IP blocks.
- **Netdiscover:** `netdiscover -i eth0` to find all devices connected to the network.
- **Website Vulnerability Scanning:**
 - [AS93 Web Check](#): Best site to find vulnerabilities.

4. Key Cybersecurity Concepts

- **Brute Force Attacks:** Use password and username lists to test combinations.
- **Load Balancer:** Distributes traffic among servers to manage loads efficiently.
- **Malware:** Malicious software or program designed to harm systems.
- **Zero-Day Exploits:** Newly discovered vulnerabilities without a patch.
- **C2 (Command and Control):** Mechanism attackers use to control compromised systems.

5. Prerequisites for Cybersecurity

- **Networking Knowledge**
- **System Administration:** Includes hardware expertise.
- **Software Development Skills**

6. Job Sectors

- **High Demand Areas:**
 - Vendors
 - Banks
 - Financial Organizations (offering higher pay than banks)
 - Other industries

7. Cybersecurity Certifications

- **CISSP (Most Demanded in the USA/Europe):** [Learn More](#)
- **CompTIA Certifications:** [Visit Site](#)
- **OSCP (Best Course Overall):** [Learn More](#)
- **Mile2 Certifications:** [Explore](#)

- **CEH (Certified Ethical Hacker):** Beginner-friendly certification from [EC-Council](#).